Model- and Data-Based Approaches to Wind Compensation for Airship Drones

Alexander Niclas Soukup

Master's Thesis

2026

Supervisor: Sebastian Gieray
M. Sc.

Examiner: Walter Fichter, Univ.-Prof. Dr.-Ing.

University of Stuttgart
Institute of Flight Mechanics and Controls

Erklärungen für Abschlussarbeiten am iFR

Erklärungen bei Ausgabe des Themas der Arbeit

Mir ist bekannt, dass die Arbeit im Online-Katalog der Bibliothek und auf der Institutswebseite erfasst wird, was eine dauerhafte, weltweite Sichtbarkeit der bibliographischen Daten der Arbeit zur Folge hat.

☐ Ich stimme der Nennung meines Namens und einer Kurzfassung der Arbeit zu diesem Zweck zu.

☐ Ich räume der Universität Stuttgart, Institut für Flugmechanik und Flugregelung, an dieser Arbeit und den im Rahmen dieser Arbeit entstandenen Ergebnissen ein kostenloses, zeitlich und räumlich unbeschränktes, einfaches Nutzungsrecht ein. Dieses erstreckt sich auf sämtliche bekannte Nutzungsarten und umfasst neben dem Recht auf Nutzung in Forschung, Lehre und Studium insbesondere das Recht der Vervielfältigung, Verbreitung und Änderung sowie das Recht der öffentlichen Zugänglichmachung.

Ort, Datum	Alexander Niclas Soukup

Erklärungen bei Abgabe der Arbeit

Hiermit versichere ich, dass ich diese Arbeit selbständig mit Unterstützung der Betreuer angefertigt und keine anderen als die angegebenen Quellen und Hilfsmittel verwendet habe. Fremde Beiträge habe ich als solche gekennzeichnet. Die Arbeit oder wesentliche Bestandteile davon sind weder an dieser noch an einer anderen Bildungseinrichtung bereits Gegenstand einer Prüfungsleistung gewesen und wurden nicht ohne Zustimmung der Prüferin bzw. des Prüfers bereits veröffentlicht. Ich erkläre weiterhin, bei der Erstellung der Arbeit die einschlägigen Bestimmungen entsprechend den Regeln guter wissenschaftlicher Praxis[1] eingehalten zu haben. Ich versichere, dass alle gedruckten Exemplare mit der digitalen Abgabe der Arbeit übereinstimmen.

Sollte ich der Universität Stuttgart, Institut für Flugmechanik und Flugregelung, Nutzungsrechte an dieser Arbeit und den im Rahmen dieser Arbeit entstandenen Ergebnissen eingeräumt haben, versichere ich, eventuell erforderlich gewordene Zustimmungen der Urheber zur Nutzung fremder Beiträge in meiner Arbeit eingeholt zu haben. Ich erkläre mich für den Fall der Verletzung der Rechte Dritter durch meine Arbeit bereit, der Universität Stuttgart einen daraus entstehenden Schaden zu ersetzen bzw. die Universität Stuttgart auf deren Aufforderung von eventuellen Ansprüchen Dritter freizustellen. Zusätzlich zu den vom Prüfer geforderten Exemplaren, übertrage ich der Universität Stuttgart das Eigentum an einem von mir der Bibliothek des Instituts für Flugmechanik und Flugregelung kostenlos zur Verfügung gestellten Exemplar dieser Arbeit.

[1] Siehe “Leitlinien zur Sicherung guter wissenschaftlicher Praxis” der DFG bzw. “Satzung der Universität Stuttgart zur Sicherung der Integrität wissenschaftlicher Praxis und zum Umgang mit Fehlverhalten in der Wissenschaft”.

Ort, Datum

Alexander Niclas Soukup

Abstract

In here, you should briefly describe the thesis content in the primary language.

Kurzfassung

In here, you should briefly describe the thesis content in the secondary language.

Acknowledgements

Contents

Erklärungen iii

Kurzfassung / Abstract v

Acknowledgements vii

List of Figures xi

List of Tables xiii

Nomenclature xv

1. Introduction 1

1.1. Problem setting . . . 1
1.2. Aim of this work . . . 1
1.3. State of the art / Related work . . . 2
1.4. Airship squid . . . 2
1.4.1. Hardware . . . 2
1.4.2. Software . . . 2
1.5. Thesis Structure . . . 3

2. Fundamentals 5

2.1. Coordinate Systems . . . 5
2.1.1. Body-Fixed Coordinate System . . . 5
2.1.2. Earth-Fixed Coordinate System . . . 5
2.2. Airship Model . . . 6
2.2.1. Gravitation and Buoyancy . . . 8
2.2.2. Aerodynamics . . . 9
2.2.3. Propulsion . . . 12
2.3. Wind Model . . . 12
2.3.1. Dryden Turbulence Model . . . 13
2.4. Kalman Filter . . . 13
2.4.1. Extended Kalman Filter . . . 13
2.4.2. Unscented Kalman Filter . . . 13
2.5. Disturbance Feedforward . . . 13

3. model-based wind estimation and compensation 15

3.1. Model . . . 15
3.1.1. Discretization . . . 15
3.2. Kalman Filter . . . 15
3.2.1. discrete state transition function . . . 15
3.2.2. discrete measurement function . . . 15
3.2.3. Process Noise and Measurement Noise . . . 16

3.3. Implementation of UKF . . . 16
3.4. Implementation of EKF . . . 16
3.4.1. Analytical vs Numerical Jacobian . . . 16
3.4.2. Robustify EKF . . . 16
3.5. Compensation . . . 16
3.5.1. Disturbance Feedforward . . . 17
3.5.2. Nonlinear Dynamic Inversion . . . 18

4. data-driven wind estimation and compensation **19**

5. Evaluation and results **21**
5.1. Wind estimation . . . 21
5.1.1. Sensitivity to wind parameters . . . 21
5.1.2. Effect of Airspeed Sensor . . . 22
5.2. Wind compensation . . . 23
5.2.1. Problems with Observability . . . 23
5.3. Flight Test . . . 24

6. Conclusion and Outlook **25**

Bibliography **27**

A. First Appendix **29**
A.1. Parameters of the airship squid . . . 29

B. Second Appendix **31**

List of Figures

2.1. Definition of the earth-fixed and body-fixed coordinate systems. 6

3.1. Block diagram of the control structure, showing the control structure with combined PID and compensation. 17

5.1. Comparison of Total RMSE Across All Estimated $W20$ Wind Speeds. The orange bar highlights the true configuration scenario where $W20_{\text{est}} = W20_{\text{act}}$. 22

5.2. Comparision of wind estimation performance for different estimated W20, W20 actual is 5. 23

List of Tables

A.1. General Airship Parameters and Specifications 29
A.2. Modeling Coefficients and constants 29

Nomenclature

Abbreviations

COG	Center of Gravity
COV	Center of Volume
CS	Coordinate System
DFF	Disturbance Feedforward
EKF	Extended Kalman Filter
KF	Kalman Filter
NDI	Nonlinear Dynamic Inversion
NED	North-East-Down convention
UKF	Unscented Kalman Filter

Latin Letters

$\boldsymbol{I}$	Inertia Matrix with respect to the COG

Greek Letters

ω	Angular velocity ($\mathrm{rad\,s^{-1}}$)

Indices

$(\cdot)_B$	Quantity describing the body
$_B(\cdot)$	Quantity in body-fixed coordinate system
$_E(\cdot)$	Quantity in Earth coordinate system

1. Introduction

This chapter serves as an overview of existing methods and ...

1.1. Problem setting

- Why wind is a challenge for airships - Why onboard wind estimation is necessary - Why compensation improves control performance

Airships are characterized by their large volume, low flight speed and high aerodynamic drag, making them considerably more susceptible to atmospheric disturbances than conventional fixed-wing aircraft. Even moderate wind speeds may become comparable to the vehicle velocity, causing significant deviations from the desired trajectory and degrading the overall control performance. This sensitivity represents one of the major challenges for autonomous airship operation, particularly during low-speed maneuvers such as hovering, station keeping, inspection tasks and precision landing.

Since the aerodynamic forces and moments acting on an airship depend on the relative velocity between the vehicle and the surrounding air, the ambient wind directly influences the vehicle dynamics. However, wind cannot usually be measured accurately using onboard sensors alone. Installing dedicated wind measurement systems increases the system complexity, weight and cost and is often impractical for small autonomous airships. Consequently, the wind must be estimated from the available onboard measurements together with a mathematical model of the vehicle dynamics.

Reliable wind estimation enables more than merely observing the surrounding atmosphere. By incorporating the estimated wind into the flight controller, the aerodynamic disturbances can be compensated before they lead to significant tracking errors. Such a model-based compensation has the potential to improve trajectory tracking, reduce control effort and increase the robustness of autonomous flight under varying environmental conditions.

Achieving accurate wind estimation and effective disturbance compensation, however, is challenging. The vehicle dynamics are highly nonlinear and are governed by the interaction of gravitational, buoyancy, propulsion and aerodynamic forces, while the aerodynamic effects themselves depend on the unknown wind. Consequently, both the estimator and the controller must account for the nonlinear system dynamics and the coupling between vehicle motion and the surrounding airflow.

1.2. Aim of this work

- Objectives - Scope - Evaluation scenarios

The aim of this work is to develop and implement a model-based algorithm for wind estimation and wind compensation for an unmanned airship. The proposed approach shall reliably estimate the ambient wind using only the available onboard measurements and the nonlinear dynamic model of the vehicle, without requiring dedicated wind sensors.

The wind estimation algorithm is designed to operate under conditions representative of realistic flight scenarios. These include dynamic wind conditions with gusts of up to 30 km/h, as well as different flight maneuvers such as forward and sideward flight, rotation about the vertical axis, and coordinated turning during forward flight. The estimated wind shall remain accurate and robust despite the nonlinear vehicle dynamics and changing operating conditions.

Based on the estimated wind, a model-based compensation strategy is implemented to directly counteract the aerodynamic disturbances acting on the airship. By incorporating the estimated wind into the control system, the influence of wind on the vehicle motion is reduced as directly as possible, thereby improving trajectory tracking performance and the overall robustness of the flight controller.

To evaluate the proposed approach, the complete estimation and compensation framework is implemented in a MATLAB/Simulink simulation environment using a comprehensive nonlinear airship model. The performance of the estimator and the effectiveness of the compensation strategy are assessed under various wind conditions and flight maneuvers representative of realistic operating scenarios.

1.3. State of the art / Related work

- Airship dynamics - Wind estimation - Wind compensation / disturbance observers - Positioning of your contribution

1.4. Airship squid

This work is performed on the airship squid, which is further specified in this section. The airship was designed and built by roboloon and consists of a elastic hull, which is usually filled with Helium. For maneuvering the airships has one stern propeller at the back, four fins and four impellers integrated in the fins totalling 9 control surfaces making the airship an overactuated system.

Picture of airship

A full table showing the specifications of the airship can be found in Table A.1

1.4.1. Hardware

The airship is equipped with an IMU, GPS and a 1D pitot tube measuring the airspeed in the body-fixes x axis. This pitot tube yields the restriction of only measuring airspeeds abve 1.5m/s, below this threshold no measurement is available.

1.4.2. Software

The airship is controlled by a PID controller ... The PID is tuned to allow for robust behavior in all flight envelopes.

All algorithms are first implemented in simulation. This simulation poses a model of the airship as derived in Chapter and an implementation of the PID controller. Additionally uncertainties for the forces are considered. After successful validation in the simulation a real world flight test is carried out.

ref

Allocator

1.5. Thesis Structure

2. Fundamentals

This chapter ...

2.1. Coordinate Systems

To describe the motion and dynamics of the airship, as well as the wind velocity acting on the vehicle, two coordinate systems (CS) are considered throughout this work: the body-fixed coordinate system and the earth-fixed coordinate system. All relevant quantities are expressed in one of these coordinate systems. Their definitions and the corresponding coordinate transformations are introduced in the following [**Fichter2009Flugmechanik**].

2.1.1. Body-Fixed Coordinate System

The body-fixed coordinate system B is attached to the airship and therefore translates and rotates together with the vehicle. Its origin is located at the Center of Gravity (COG) of the airship, and the coordinate axes are aligned with the body axes according to the North-East-Down (NED) convention. Consequently, the x_B-axis points towards the nose of the airship, the y_B-axis points towards the right side, and the z_B-axis points downwards.

The body-fixed coordinate system is used to describe quantities directly related to the vehicle dynamics, such as translational and rotational velocities, aerodynamic forces, propulsion forces, and moments. Since the equations of motion are formulated in the body-fixed frame, this coordinate system represents the primary reference frame for the dynamic model. A quantity expressed in the body-fixed coordinate system is indicated by a preceding frame identifier, denoted as $_B(\cdot)$.

add picture of airship and its CS

2.1.2. Earth-Fixed Coordinate System

The earth-fixed coordinate system E is introduced as an inertial reference frame for describing the position and orientation of the airship, as well as the wind velocity. For simplicity, a flat-earth assumption is made, which neglects the curvature and rotation of the Earth. This assumption is justified by the relatively short flight distances considered in this work.

The origin of the earth-fixed coordinate system is chosen as an arbitrary point on the Earth's surface. The axes are aligned according to the NED convention, where the north and east axes span the horizontal plane and the down axis points towards the Earth's center. Since the earth-fixed coordinate system is assumed to be inertial, it is considered non-accelerating and non-rotating. A quantity expressed in the earth-fixed coordinate system is indicated by a preceding frame identifier, denoted as $_E(\cdot)$.

The transformation from the earth-fixed coordinate system to the body-fixed coordinate system is described by the rotation matrix [1]

$$\boldsymbol{R}_{BE} = \begin{bmatrix} c_\psi c_\theta & s_\psi c_\theta & -s_\theta \\ c_\psi s_\theta s_\phi - s_\psi c_\phi & s_\psi s_\theta s_\phi + c_\psi c_\phi & c_\theta s_\phi \\ c_\psi s_\theta c_\phi + s_\psi s_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi & c_\theta c_\phi \end{bmatrix}, \tag{2.1}$$

which maps vectors expressed in the earth-fixed coordinate system into the body-fixed coordinate system, where $s_{(\cdot)}$ and $c_{(\cdot)}$, denote $\sin(\cdot)$ and $\cos(\cdot)$. The inverse transformation is given by the transpose of the rotation matrix

$$\boldsymbol{R}_{EB} = (\boldsymbol{R}_{BE})^T \,. \tag{2.2}$$

Figure 2.1.: Definition of the earth-fixed and body-fixed coordinate systems.

2.2. Airship Model

The airship is modeled using a six degree-of-freedom (6DOF) Newton Euler approach [1]. Therefore we assume the airship to be a rigid body with constant mass and model the translational and rotational dynamics accordingly. We express the equations of motion in body-coordinates, allowing for a constant inertia matrix. Unless stated otherwise, all quantities are expressed in the body-fixed coordinate frame, and all forces and moments are assumed to act at the center of gravity (COG). The corresponding frame identifier ${}_B(\cdot)$ and point-of-application identifier $(\cdot)^{\mathrm{COG}}$ are omitted throughout the remainder of this section for improved readability.

The three-dimensional **translational dynamics** in body-coordinates are given by the linear momentum theorem as

$$m\boldsymbol{I}_{3\times3}\,\boldsymbol{a}_B = \boldsymbol{F}_{\mathrm{ext}}, \tag{2.3}$$

where m denotes the mass of the airship, $\boldsymbol{I}_{3\times3} \in \mathbb{R}^{3\times3}$ denotes the identity matrix, $\boldsymbol{a}_B \in \mathbb{R}^3$ describes the acceleration of the body (airship) in body-fixed coordinates and $\boldsymbol{F}_{\mathrm{ext}} \in \mathbb{R}^3$ represents the resultant external force acting at the center of gravity (COG), expressed in body-fixed coordinates.

To relate the body acceleration $\boldsymbol{a}_B$ to the body velocity $\boldsymbol{v}_B \in \mathbb{R}^3$, the rotation of the body-fixed coordinate system relative to the earth-fixed coordinate system must be taken into account. Applying Euler's transport theorem yields

$$\boldsymbol{a}_B = \left(\frac{d}{dt}\boldsymbol{v}_B\right) = \dot{\boldsymbol{v}}_B + (\boldsymbol{\omega}_{EB} \times \boldsymbol{v}_B) = \dot{\boldsymbol{v}}_B + \boldsymbol{a}_{\mathrm{cor}}, \tag{2.4}$$

where $\boldsymbol{a}_{\mathrm{cor}} \in \mathbb{R}^3$ denotes the Coriolis acceleration as a result of the rotating body-fixed CS and $\boldsymbol{\omega}_{EB} \in \mathbb{R}^3$ describes the rotational velocity of the body-fixed CS against the Earth CS. As the Earth CS is inertial this reduces to the angular velocity of the airship along its three axes, $\boldsymbol{\omega}_B \in \mathbb{R}^3$. Inserting equation (2.4) in equation (2.3) and defining the Coriolis force as

$$\boldsymbol{F}_{\mathrm{cor}} = m\boldsymbol{I}_{3\times3}\,\boldsymbol{a}_{\mathrm{cor}}, \tag{2.5}$$

yields the dynamics for the translational motion of the airship in body-fixed coordinates

$$m\boldsymbol{I}_{3\times 3}\,\dot{\boldsymbol{v}}_B = \boldsymbol{F}_{\text{ext}} - \boldsymbol{F}_{\text{cor}}. \tag{2.6}$$

The three-dimensional **rotational dynamics** in body-fixed coordinates are given by the angular momentum theorem as

$$\left(\frac{d}{dt}\left(\boldsymbol{I}_B\,\boldsymbol{\omega}_B\right)\right) = \boldsymbol{M}_{\text{ext}}, \tag{2.7}$$

where $\boldsymbol{I}_B \in \mathbb{R}^{3\times 3}$ denotes the inertia matrix of the airship expressed in body-fixed coordinates and $\boldsymbol{M}_{\text{ext}} \in \mathbb{R}^3$ represents the resultant external moment acting about the center of gravity (COG), expressed in body-fixed coordinates.

To relate the change of angular momentum to the angular acceleration of the airship, the rotation of the body-fixed coordinate system relative to the earth-fixed coordinate system must be considered. Applying Euler's transport theorem to the angular momentum vector yields

$$\left(\frac{d}{dt}\left(\boldsymbol{I}_B\,\boldsymbol{\omega}_B\right)\right) = \boldsymbol{I}_B\,\dot{\boldsymbol{\omega}}_B + \left(\boldsymbol{\omega}_B \times \boldsymbol{I}_B\,\boldsymbol{\omega}_B\right), \tag{2.8}$$

where the second term represents the gyroscopic acceleration caused by the rotation of the body-fixed coordinate system. Since the inertia matrix is constant when expressed in body-fixed coordinates, the time derivative only applies to the angular velocity.

Defining the gyroscopic moment as

$$\boldsymbol{M}_{\text{gyro}} = \left(\boldsymbol{\omega}_B \times \boldsymbol{I}_B\,\boldsymbol{\omega}_B\right) \tag{2.9}$$

and rearranging equation (2.7) yields the rotational dynamics of the airship in body-fixed coordinates

$$\boldsymbol{I}_B\,\dot{\boldsymbol{\omega}}_B = \boldsymbol{M}_{\text{ext}} - \boldsymbol{M}_{\text{gyro}}. \tag{2.10}$$

We can combine the translational dynamics (2.6) and the rotational dynamics (2.10) into one equation obtaining the Newton Euler Equations for the airship.

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{ext}} - \boldsymbol{\mathcal{F}}_{\text{cor}} \tag{2.11}$$

with the generalized velocity vector $\boldsymbol{\nu}$ and mass matrix $\boldsymbol{M}$ defined as

$$\boldsymbol{\nu} = \begin{bmatrix}\boldsymbol{v}_B\\ \boldsymbol{\omega}_B\end{bmatrix} \in \mathbb{R}^6, \qquad \boldsymbol{M} = \begin{bmatrix} m\boldsymbol{I}_{3\times 3} & \boldsymbol{0}_{3\times 3}\\ \boldsymbol{0}_{3\times 3} & \boldsymbol{I}_B\end{bmatrix} \in \mathbb{R}^{6\times 6}. \tag{2.12}$$

The generalized external and Coriolis force vectors are defined as

$$\boldsymbol{\mathcal{F}}_{\text{ext}} = \begin{bmatrix}\boldsymbol{F}_{\text{ext}}\\ \boldsymbol{M}_{\text{ext}}\end{bmatrix} \in \mathbb{R}^6, \qquad \boldsymbol{\mathcal{F}}_{\text{cor}} = \begin{bmatrix}\boldsymbol{F}_{\text{cor}}\\ \boldsymbol{M}_{\text{cor}}\end{bmatrix} \in \mathbb{R}^6. \tag{2.13}$$

The generalized external forces acting on the airship can be further classified based on the available knowledge of the airship system. Since the airship is modeled as a rigid body, the internal deformations and distributed effects of the structure are neglected, which represents a significant simplification of the real system. Consequently, the accuracy of the airship model depends strongly on the precise representation of the external forces and moments acting on the vehicle. To capture the relevant physical behavior of the airship, these contributions must therefore be modeled as accurately as possible.

However, due to the complexity of the real environment and the limitations of the available model knowledge, not all forces and moments acting on the airship can be fully accounted for. The resulting discrepancies between modeled and actual system behavior are considered model uncertainties. These uncertainties are not explicitly modeled within the presented equations of motion but are addressed in the subsequent estimation and control design.

The external forces and moments acting on the airship are divided into four main categories: gravitational, buoyancy, aerodynamic, and propulsion.

$$\boldsymbol{\mathcal{F}}_{\text{ext}} = \boldsymbol{\mathcal{F}}_{\text{grav}} + \boldsymbol{\mathcal{F}}_{\text{buoy}} + \boldsymbol{\mathcal{F}}_{\text{aero}} + \boldsymbol{\mathcal{F}}_{\text{prop}}, \tag{2.14}$$

resulting in the final EOM

$$\boldsymbol{M}\dot{\boldsymbol{\nu}} = \boldsymbol{\mathcal{F}}_{\text{grav}} + \boldsymbol{\mathcal{F}}_{\text{buoy}} + \boldsymbol{\mathcal{F}}_{\text{aero}} + \boldsymbol{\mathcal{F}}_{\text{prop}} - \boldsymbol{\mathcal{F}}_{\text{cor}}. \tag{2.15}$$

The generalized external forces are derived in the following.

2.2.1. Gravitation and Buoyancy

The gravitational force computes to

$$\boldsymbol{F}_{\text{grav}} = m\boldsymbol{g} = m\boldsymbol{R}_{BE}\,{}_{E}\boldsymbol{g} \qquad \text{with} \qquad {}_{E}\boldsymbol{g} = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}, \tag{2.16}$$

where ${}_{E}\boldsymbol{g}$ represents the constant gravity vector in the inertial earth reference frame following the NED convention and $\boldsymbol{R}_{BE}$ denotes the transformation matrix from earth CS to body CS.

As the gravitational force acts at the center of gravity, no gravitational moment about the COG is induced, thus

$$\boldsymbol{\mathcal{F}}_{\text{grav}} = \begin{bmatrix} \boldsymbol{F}_{\text{grav}} \\ \boldsymbol{0}_{3\times 1} \end{bmatrix}. \tag{2.17}$$

The buoyancy depends on the density of the surrounding air ρ_{air} and the lifting gas contained within the hull ρ_{gas} as well as the volume of the airship V. Assuming a homogeneous distribution, the resulting buoyancy force acts at the center of volume (COV) and is given by

$$\boldsymbol{F}_{\text{buoy}}^{\text{COV}} = -V\boldsymbol{g}(\rho_{\text{air}} - \rho_{\text{gas}}). \tag{2.18}$$

In case of the airship *Squid*, the volume is chosen such that $\boldsymbol{F}_{\text{grav}} + \boldsymbol{F}_{\text{buoy}} > \boldsymbol{0}$, resulting in a downward-directed net force that prevents the airship from ascending without external input. To express the buoyancy contribution with respect to the center of gravity, the moment induced by the offset between the COV and COG is considered:

$$\boldsymbol{M}_{\text{buoy}} = \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{buoy}}^{\text{COV}}, \tag{2.19}$$

where $\boldsymbol{r}_{\mathrm{COV}} \in \mathbb{R}^3$ denotes the vector from the COG to COV. Due to the design of the airship, $\boldsymbol{r}_{\mathrm{COV}}$ only contains a component along the z_B-direction. Consequently, the generalized buoyant force expressed at the COG is given by

$$\boldsymbol{\mathcal{F}}_{\mathrm{buoy}} = \begin{bmatrix} \boldsymbol{F}_{\mathrm{buoy}} \\ \boldsymbol{M}_{\mathrm{buoy}} \end{bmatrix} \qquad \text{with} \qquad \boldsymbol{F}_{\mathrm{buoy}} = \boldsymbol{F}_{\mathrm{buoy}}^{\mathrm{COV}}. \tag{2.20}$$

2.2.2. Aerodynamics

The aerodynamic forces and moments play a central role in the airship dynamics, as they directly depend on the aerodynamic velocity. The aerodynamic velocity is defined as

$$\boldsymbol{v}_A = \boldsymbol{v}_B - \boldsymbol{v}_W \tag{2.21}$$

and denotes the relative velocity between the airship and the surrounding air (often called *airspeed*), where $\boldsymbol{v}_W \in \mathbb{R}^3$ corresponds to the wind velocity expressed in body-fixed coordinates. As a result, the aerodynamical forces and moments contain information about the current wind conditions. An accurate aerodynamic model is therefore essential and forms the basis for the wind estimation approach presented in this work.

As described in [2], the aerodynamic forces and moments can be classified into several contributions, including drag, added mass, viscous effects, and fin forces. Since these contributions account for the majority of the aerodynamic effects, they are considered in the present model.

$$\boldsymbol{\mathcal{F}}_{\mathrm{aero}} = \boldsymbol{\mathcal{F}}_{\mathrm{drag}} + \boldsymbol{\mathcal{F}}_{\mathrm{am}} + \boldsymbol{\mathcal{F}}_{\mathrm{visc}} + \boldsymbol{\mathcal{F}}_{\mathrm{fins}} \tag{2.22}$$

A detailed derivation of the individual contributions is omitted here; instead, the resulting expressions are summarized for completeness. An extensive derivation of the aerodynamic forces and moments can be found in [2]. The aerodynamic forces act at the center of volume. Therefore, the offset between the COV and the COG results in an induced moment, which is considered in the aerodynamic moment contribution.

Aerodynamic Drag

The aerodynamic drag force opposes the relative motion between the airship and the surrounding air and therefore acts in the opposite direction of the aerodynamic velocity. In this work, the drag force is modeled independently along each body-fixed axis using a quadratic drag formulation based on the conventional drag equation [3]. The resulting aerodynamic drag force is given by

$$\boldsymbol{F}_{\mathrm{drag}}^{\mathrm{COV}} = -\frac{1}{2}\rho_{\mathrm{air}}\,\mathrm{sgn}(\boldsymbol{v}_A) \odot \boldsymbol{v}_A^2 \odot \begin{bmatrix} c_{\mathrm{D,ax}} \\ c_{\mathrm{D,lat}} \\ c_{\mathrm{D,lat}} \end{bmatrix}, \tag{2.23}$$

where $\odot$ represents the element-wise (Hadamard) product, and $\mathrm{sgn}(\cdot)$ is applied element-wise. The parameters $c_{\mathrm{D,ax}}$ and $c_{\mathrm{D,lat}}$ are the axial and lateral drag coefficients, respectively. Since the airship is assumed to be rotationally symmetric about its longitudinal axis, the same drag coefficient is used for the lateral and vertical directions.

Since the aerodynamic drag force acts at the COV, it must be transformed to the COG to obtain the corresponding generalized aerodynamic force as

$$\boldsymbol{\mathcal{F}}_{\text{drag}} = \begin{bmatrix} \boldsymbol{F}_{\text{drag}} \\ \boldsymbol{M}_{\text{drag}} \end{bmatrix} \quad \text{with} \quad \boldsymbol{M}_{\text{drag}} = \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{drag}}^{\text{COV}}, \quad \boldsymbol{F}_{\text{drag}} = \boldsymbol{F}_{\text{drag}}^{\text{COV}}. \tag{2.24}$$

Added Mass

For an airship, the added mass can be interpreted as the effective mass of the surrounding air that is accelerated together with the vehicle [4]. The associated added-mass forces and moments arise from the inertia of this accelerated air and contribute to the equations of motion in two ways: through a constant added-mass matrix and through a velocity-dependent coupling term:

$$\boldsymbol{\mathcal{F}}_{\text{am}}^{\text{COV}} = -\begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} \\ \boldsymbol{M}_{21} & \boldsymbol{M}_{22} \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{v}}_B \\ \dot{\boldsymbol{\omega}}_B \end{bmatrix} - \begin{bmatrix} \boldsymbol{\omega}_B \times (\boldsymbol{M}_{11}\boldsymbol{v}_A + \boldsymbol{M}_{12}\boldsymbol{\omega}_B) \\ \boldsymbol{v}_A \times (\boldsymbol{M}_{11}\boldsymbol{v}_A + \boldsymbol{M}_{12}\boldsymbol{\omega}_B) + \boldsymbol{\omega}_B \times (\boldsymbol{M}_{21}\boldsymbol{v}_A + \boldsymbol{M}_{22}\boldsymbol{\omega}_B) \end{bmatrix} \tag{2.25}$$

For steady translation ($\boldsymbol{\omega}_B = \boldsymbol{0}$), the coupling term reduces to the classical *Munk moment*, $-\boldsymbol{v}_A \times (\boldsymbol{M}_{11}\boldsymbol{v}_A)$, which is a destabilizing moment characteristic of elongated bodies.

The entries of the added-mass matrix are computed using the analytical expressions given by [2] and are listed in Appendix .

add ref and write M in appendix

Since the generalized added-mass force acts at the COV, it is transformed to the COG as

$$\boldsymbol{\mathcal{F}}_{\text{am}} = \begin{bmatrix} \boldsymbol{F}_{\text{am}} \\ \boldsymbol{M}_{\text{am}}^{\text{COV}} + \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{am}} \end{bmatrix} \quad \text{with} \quad \boldsymbol{F}_{\text{am}} = \boldsymbol{F}_{\text{am}}^{\text{COV}}. \tag{2.26}$$

Viscous Aerodynamic Effects

In contrast to the added-mass forces, which are associated with the acceleration of the surrounding air, the viscous forces and moments originate from the interaction between the airship surface and the surrounding airflow. These effects arise from viscous shear stresses and pressure differences and are described in detail by [3].

The magnitude of the viscous force and moment acting normal to the hull centerline are given by

$$F_{\text{N,visc}}^{\text{COV}} = q_0 \left(-C_{\text{F,hif}} \sin 2\gamma + C_{\text{F,hvf}} \sin^2 \gamma\right) \tag{2.27}$$

$$M_{\text{N,visc}}^{\text{COV}} = q_0 \left(-C_{\text{M,hif}} \sin 2\gamma + C_{\text{M,hvf}} \sin^2 \gamma\right), \tag{2.28}$$

where q_0 describes the dynamic pressure associated by the local aerodynamic velocity $\boldsymbol{v}_{A,0}$ and γ denotes the angle between the hull centerline and $\boldsymbol{v}_{A,0}$:

$$q_0 = \frac{1}{2}\rho_{\text{air}} \left\|\boldsymbol{v}_{A,0}\right\|_2^2 \quad \text{and} \quad \gamma = \text{atan2}\left(\sqrt{v_{A,0}^2 + w_{A,0}^2}/u_{A,0}\right). \tag{2.29}$$

Here $\boldsymbol{v}_{A,0} = [u_{A,0},\, v_{A,0},\, w_{A,0}]^T$ denotes the aerodynamic velocity at ε_0, which marks the point along the hull, measured from the nose, beyond which correction for real flow behavior is necessary. The coefficient $C_{\cdot,\text{hif}}$ accounts for the additional drag caused by the pressure difference between the front and rear of the hull after flow separation, whereas $C_{\cdot,\text{hvf}}$ represents the crossflow drag caused by the separated flow, modeled using a cylinder-drag analogy.

The viscous force and moment are then decomposed into their body-axis components. Since both act normal to the hull centerline $F^{\mathrm{COV}}_{\mathrm{visc},x} = M^{\mathrm{COV}}_{\mathrm{visc},x} = 0$ and

$$F^{\mathrm{COV}}_{\mathrm{visc},y} = -F^{\mathrm{COV}}_{\mathrm{N,visc}} \frac{v_{A,0}}{\sqrt{v^2_{A,0} + w^2_{A,0}}}, \qquad F^{\mathrm{COV}}_{\mathrm{visc},z} = -F^{\mathrm{COV}}_{\mathrm{N,visc}} \frac{w_{A,0}}{\sqrt{v^2_{A,0} + w^2_{A,0}}} \tag{2.30}$$

and the corresponding body-axis moment components are

$$M^{\mathrm{COV}}_{\mathrm{visc},y} = M^{\mathrm{COV}}_{\mathrm{N,visc}} \frac{v_{A,0}}{\sqrt{v^2_{A,0} + w^2_{A,0}}}, \qquad M^{\mathrm{COV}}_{\mathrm{visc},z} = -M^{\mathrm{COV}}_{\mathrm{N,visc}} \frac{w_{A,0}}{\sqrt{v^2_{A,0} + w^2_{A,0}}}. \tag{2.31}$$

Again, we have to transform the force and moment to the COG.

$$\boldsymbol{\mathcal{F}}_{\mathrm{visc}} = \begin{bmatrix} \boldsymbol{F}_{\mathrm{visc}} \\ \boldsymbol{M}^{\mathrm{COV}}_{\mathrm{visc}} + \boldsymbol{r}_{\mathrm{COV}} \times \boldsymbol{F}_{\mathrm{visc}} \end{bmatrix} \qquad \text{with} \qquad \boldsymbol{F}_{\mathrm{visc}} = \boldsymbol{F}^{\mathrm{COV}}_{\mathrm{visc}}. \tag{2.32}$$

Fin Forces

The fin forces are computed using a lumped approach, in which each of the four fins is treated as a single lifting surface. The contribution of each fin depends on the local aerodynamic velocity and the corresponding angle of attack. The total fin force and moment are then obtained by summing the individual fin contributions.

For each fin $i = 1, \ldots, 4$, located at position $\boldsymbol{r}_{\mathrm{fin},i}$ relative to the COV, the local aerodynamic velocity follows from the rigid-body velocity field

$$\boldsymbol{v}_{A,\mathrm{fin},i} = \boldsymbol{v}_A + \boldsymbol{\omega}_B \times \boldsymbol{r}_{\mathrm{fin},i}, \tag{2.33}$$

from which the local dynamic pressure is obtained as

$$q_{0,\mathrm{fin},i} = \frac{1}{2} \rho_{\mathrm{air}} \left\| \boldsymbol{v}_{A,\mathrm{fin},i} \right\|_2^2. \tag{2.34}$$

The geometric angle of attack of each fin is computed from the local velocity component perpendicular to the fin surface $v_{\perp,\mathrm{fin},i}$ and the local velocity component along the hull centerline $u_{\mathrm{fin},i}$

$$\alpha_{\mathrm{fin},i} = \mathrm{atan2}\left(\frac{v_{\perp,\mathrm{fin},i}}{u_{\mathrm{fin},i}} \right), \tag{2.35}$$

and is saturated at the stall angle α_{stall}.

The resulting normal force on each fin computes to

$$F_{N,\mathrm{fin},i} = -q_{0,\mathrm{fin},i}\, C_{\mathrm{F,fnf}}\, \alpha_{\mathrm{fin},i}, \tag{2.36}$$

where $C_{\mathrm{F,fnf}}$ is the empirical fin normal-force coefficient that is assumed to capture all aerodynamic effects of the fins, including geometric and interference effects, in a single fitted parameter.

The total fin force and moment about the COV are obtained by summing the individual fin contributions,

$$\boldsymbol{F}_{\text{fins}}^{\text{COV}} = \sum_{i=1}^{4} \boldsymbol{F}_{\text{fin},i}, \qquad \boldsymbol{M}_{\text{fins}}^{\text{COV}} = \sum_{i=1}^{4} \boldsymbol{r}_{\text{fin},i} \times \boldsymbol{F}_{\text{fin},i}, \tag{2.37}$$

where $\boldsymbol{F}_{\text{fin},i}$ denotes the fin force vector obtained by resolving the fin normal force $F_{N,Fi}$ into its body-fixed components according to the orientation of the respective fin.

As with the previous contributions, the resulting force and moment are transformed to the COG to obtain the generalized fin forces

$$\boldsymbol{\mathcal{F}}_{\text{fins}} = \begin{bmatrix} \boldsymbol{F}_{\text{fins}}^{\text{COV}} \\ \boldsymbol{M}_{\text{fins}}^{\text{COV}} + \boldsymbol{r}_{\text{COV}} \times \boldsymbol{F}_{\text{fins}}^{\text{COV}} \end{bmatrix} \qquad \text{with} \qquad \boldsymbol{F}_{\text{fins}} = \boldsymbol{F}_{\text{fins}}^{\text{COV}}. \tag{2.38}$$

2.2.3. Propulsion

The propulsion forces and moments resulting from the command input $\boldsymbol{u} \in \mathbb{R}^9$ are computed in two steps. First, the thrust generated by each individual motor $F_{\text{motor},i}$ is determined from the corresponding motor command u_i using a nonlinear motor model. Due to the different characteristics of the impellers and the stern propeller, separate parameter sets are used for the two actuator types. The resulting motor thrust is therefore given by

$$F_{\text{motor},i} = \begin{cases} \alpha_{\text{imp}} \left(k_{\text{imp}}\, u_i^2 + (1 - k_{\text{imp}})u_i\right), & i = 1, \dots, 8 \\ \alpha_{\text{stern}} \operatorname{sgn}(u_i) \left(k_{\text{stern}}\, u_i^2 + (1 - k_{\text{stern}})u_i\right), & i = 9 \end{cases} \tag{2.39}$$

The individual motor thrusts are then combined into the resulting generalized force vector by considering the orientation and position of each motor. This transformation is described by the motor effectiveness matrix $\boldsymbol{B}_{\text{eff}}^{\text{COG}} \in \mathbb{R}^{6\times 9}$, which maps the individual motor thrusts to the generalized force vector acting at the COG. The matrix incorporates the orientation of each motor as well as the moment contributions resulting from their respective positions relative to the COG.

$$\boldsymbol{\mathcal{F}}_{\text{prop}} = \boldsymbol{B}_{\text{eff}}^{\text{COG}} \boldsymbol{F}_{\text{motor}}, \tag{2.40}$$

with

$$\boldsymbol{B}_{\text{eff}}^{\text{COG}} = \begin{bmatrix} \boldsymbol{d}_1 & \cdots & \boldsymbol{d}_9 \\ \boldsymbol{r}_1 \times \boldsymbol{d}_1 & \cdots & \boldsymbol{r}_9 \times \boldsymbol{d}_9 \end{bmatrix}, \tag{2.41}$$

where $\boldsymbol{r}_i \in \mathbb{R}^3$ denotes the position vector of the i-th motor relative to the COG, and $\boldsymbol{d}_i \in \mathbb{R}^3$ denotes the corresponding unit thrust direction vector.

2.3. Wind Model

To be able to augment our system model with a wind states we need a wind model. To make use of these dynamics an accurate wind model is desired.

Although the Van Karman Model describes the turbulence spectrum more accurately in most cases this work implements the Dryden Turbulence Model due to its easier implementation. As it results in a linear model. Van Karman is not realizable.

2.3.1. Dryden Turbulence Model

The Dryden wind model can be understood as a random walk with memory. . Inputs of the Dryden wind model is white noise. State transition functions

Add comparision plot Random walk and Dryden with appropriate parameter values

Parameters

The Dryden Model is specified by the following parameters, W20, wind direction and L and sigmas. In the 1970s the US Army developed methods to specify parameters for the Wind model. Therefore we can directly relate the L and sigmas to the current flight state according to following equations:

as W20 and wind direction are fully determined by the wind environment these cannot be easily related to the current flight status.

2.4. Kalman Filter

Kalman Filters came up in the 19xxs as a method to ... Before that state estimation methods like observers were existent. The first one was the widely popular Luenberger Observer... Kalman Filters offer the advantage to ... they also work especially well in combination with white noise inputs.

Bayesian Filtering

There exist several methods to extend the KF framework to nonlinear systems. 2 of these are explained further in the following section.

2.4.1. Extended Kalman Filter

2.4.2. Unscented Kalman Filter

2.5. Disturbance Feedforward

Disturbance Feedforward describes a method to compensate the influence of a known disturbance on the system. Linear Case Nonlinear Case

3. model-based wind estimation and compensation

There exist several estimation methods in control theory. Beginning from simple linear models like the Luenberger Observer etc... The idea is to combine the knowledge about the nonlinear airships dynamics, which have been developed in Section 2.2 and the wind model into use a state estimation method. Therefore the Kalman Filter proves to be the perfect solution optimally combining dynamics and measurements. why KF

3.1. Model

First a complete model is required to calculate the state transition function from it. Therefore we augment the six state airship model including any states that have known dynamics and are measured (position + attitude) and the wind states.

Introduce state vector (12 states) and show that all gen forces can be expressed depending on those states.

State space realization of the wind model. The parameters of the dryden wind model depend on the flight states, so we can easily express them depending on the state. Only the computation of W20 and wind direction is not state dependent, this is evaluated in ref

3.1.1. Discretization

exact discretization of the wind dynamics, approx of model dynamics

3.2. Kalman Filter

To get an Kalman Filter running we need a system in the following form

- discrete state transition function
- discrete measurement function
- Process noise
- Measurement noise

write down eq, with f, h, Q, R

We consider the case of additive process noise, which becomes clear later as we derive the equations.

3.2.1. discrete state transition function

3.2.2. discrete measurement function

The measurement function directly follows from the sensor equipment on board of the airship

3.2.3. Process Noise and Measurement Noise

we have $Q = [Q_m odel, Q_w ind]$ uncertainty through unmodeled forces uncertainty coming from wind -> discretization of covariance

measurement noise from sensor specifications Pitot tube only measurements if $u > 2m/s$. To account for this we implement a time varying measurement noise, setting $R_{pitottube} \rightarrow \infty$ if airspeed in x-direction is < 2. To understand why this works lets look at the computation of the kalman gain. R -> infinity simply means that we fully trust our dynamics and basically do not use the measurement.

RUKU vs Euler

3.3. Implementation of UKF

papers

The UKF obeys 3 Parameters, which mainly influence the spread of the sigma points. Parameters are set to beta = 2, kappa = 0 . The optimal value for alpha is determined by an optimization run.

3.4. Implementation of EKF

3.4.1. Analytical vs Numerical Jacobian

3.4.2. Robustify EKF

...

The EKF is not as robust as the UKF due to several reasons . Robustness is a important desired property of the wind estimation algorithm for the specific use case in this thesis, which is to use it for an wind compensation algorithm. Thus to further improve the robustness an Joseph Form computation of the updated covariance is implemented.

3.5. Compensation

While wind is modeled as an additional system state in the Kalman Filter implementation, it can also be considered as an external disturbance acting on the airship. Therefore, the wind estimation provided by the Kalman Filter can be utilized to compensate for the effect of wind disturbances. Two compensation approaches are implemented and evaluated: a nonlinear Disturbance Feedforward (DFF) controller and a Nonlinear Dynamic Inversion (NDI) controller. Both approaches can be interpreted as an augmentation to the existing PID controller, adding an additional compensation term

$$\boldsymbol{u} = \boldsymbol{u}_{\text{PID}} + \boldsymbol{u}_{\text{comp}}, \tag{3.1}$$

where $\boldsymbol{u}_{\text{PID}}$ is mainly responsible for setpoint tracking and stabilization, while $\boldsymbol{u}_{\text{comp}}$ uses model knowledge of the airship to counteract the wind-induced disturbances in case of DFF, or compensates for all acting forces and moments in case of the NDI.

To minimize the delay between wind disturbance estimation and compensation, the estimated disturbance is compensated directly at the force and moment level.Based on the estimated wind and the current airship state measurement, the required compensation force and moment vector

$$\boldsymbol{\mathcal{F}}_{\text{comp}} = \begin{pmatrix} \boldsymbol{F}_{\text{comp}} \\ \boldsymbol{M}_{\text{comp}} \end{pmatrix} \in \mathbb{R}^6$$

is calculated. The computation of $\mathcal{F}_{\text{comp}}$ depends on the compensation approach and is derived in the sections 3.5.1 and 3.5.2.

The total desired force and moment vector is obtained by combining the force and moment vector generated by the PID controller, $\mathcal{F}_{\text{PID}}$, with the compensation vector, $\mathcal{F}_{\text{comp}}$:

$$\mathcal{F} = \mathcal{F}_{\text{PID}} + \mathcal{F}_{\text{comp}}. \tag{3.2}$$

This force and moment vector is subsequently converted into individual motor commands by the control allocation algorithm. Performing the compensation directly at the force and moment level enables computationally efficient execution, as the mapping from forces and moments to motor commands $\boldsymbol{u}$ is performed only once per time step. At the same time, the underlying concept introduced in Eq. (3.1) is preserved, allowing disturbances to be compensated with a delay of only one control cycle.

The resulting control structure, including the compensation strategy, is illustrated in Figure 3.1.

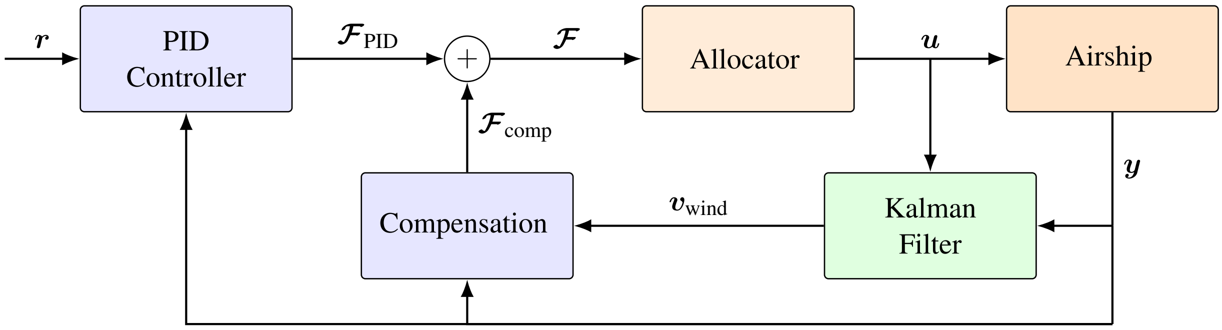

Figure 3.1.: Block diagram of the control structure, showing the control structure with combined PID and compensation.

The original PID controller, without compensation, was designed with a strong emphasis on robustness against modeling uncertainties and wind disturbances. Since the compensation approaches explicitly account for a significant portion of these disturbances, the controller can theoretically be retuned to place greater emphasis on tracking performance. This results in a more aggressively tuned PID controller. In section the effects are evaluated.

ref

In the following sections, the two compensation approaches are described in detail, and the derivation of the compensation vector $\mathcal{F}_{\text{comp}}$ is presented.

3.5.1. Disturbance Feedforward

The Disturbance Feedforward compensates for the wind induced forces and moments only. This means $\mathcal{F}_{comp}$ compensates for the additional forces and Moments created by the prevailing wind. Looking at the dynamics one can see that wind affects our airship only via the aerodynamical forces, as they depend on the airspeed Va which is

$$V_a = V_b - V_w \tag{3.3}$$

add dynamics, what is Faero and Fprop?

Fprop describes the propulsion force that is achieved by a command u. Faero is the aerodynamical force. In general Faero and Fprop describe nonlinear functions mapping u and Va to Forces. Fprop can be directly expressed by $\mathcal{F}$

we basically have everything expressed in forces.

The Disturbance Feedforward task is to find u_{comp} s.t.

$$F_{\text{aero}}(V_A) + F_{\text{prop}}(u + u_c) = F_{\text{aero}}(V_B) + F_{\text{prop}}(u) \tag{3.4}$$

To further understand what this means lets look how this unfolds assuming a linear system.

$$\begin{aligned} F(V_A) + F_{\text{prop}}(u + u_c) &= F(V_B) - F(V_w) + F_{\text{prop}}(u) + F_{\text{prop}}(u_c) && (3.5) \\ &= F(V_B) + F_{\text{prop}}(u), && (3.6) \\ \Rightarrow \quad F_{\text{prop}}(u_c) &= F(V_w) && (3.7) \end{aligned}$$

in this case F(vw) simply describes the additional forces induced by wind. For nonlinear systems its more complicated.

Tuning: The PID controller is retuned on the undisturbed model, as the compensation should eliminate those, allowing for a faster PID control. In the performance of the retuned PID controller is examined and compared to the control structure with only the PID. It shows that the retuned PID is far too aggressive and not able to achieve good performance for all flight envelopes, although performing better on some. The reason for this is that although theoretically the Compensation should compensate for the wind, it is not able to fully cancel it due to uncertainties in the wind estimation and state measurements. Hence a robust controller is desired anyways. A combination of the robust PID with the compensation is beneficial.

ref

3.5.2. Nonlinear Dynamic Inversion

As second approach a Nonlinear Dynamic Inversion (NDI) controller is implemented. The idea behind the NDI is to fully compensate the Forces acting on the Airship, basically inverting the dynamics.

$$todo \tag{3.8}$$

For NDi the model was tuned on the unforced model (so total forces were set to zero). This allows for an pretty fast PID. But again under real conditions this does not provide good results. As there is too much uncertainty in it.

Given perfect measurements and wind estimate the NDI achieves perfect compensation of all acting forces.

4. data-driven wind estimation and compensation

5. Evaluation and results

First the Performance of the Wind estimation algorithm is assessed in simulation. Later the combination of wind estimation and compensation is compared to the flight envelope without compensation. To evaluate the performance of the Estimation and compensation following flight scenarios are considered

- Forward Flight
  - constant velocity
  - acceleration
- Sideway Flight
  - constant velocity
  - acceleration
- Pure rotation areound the yaw axis
- turn while flying forward (Vorgabe von Vorwärtsgeschwindigkeit und Drehgeschwindigkeit)

5.1. Wind estimation

Following the wind estimation performance is analyzed. First of all we need to find a W20 parameter. We choose W20 = 5, which is justified in the section . ref

Result: Good wind estimation....

5.1.1. Sensitivity to wind parameters

The wind parameters W20 and wind direction cannot be measured or computed directly from the states and thus need to be estimated. In the following the sensitivity to the correct parameter is assessed. Therefore multiple tests are carried out.

First a constant parameter value for W20 was specified, creating the actual wind turbulences in the simulation. The wind estimation performance was assessed for varying estimated W20, reaching from W20 = 0 up to W20 = 30 km/h. For each estimated W20 parameter value 27 wind scenarios are run. In each wind scenario the constant wind, as well as the random seed creating the turbulences, were varied resulting in unique wind conditions for each test. For each test the RMSE between the estimated wind and actual wind are computed individually and the average RMSE along all three axis is computed. Then the results are averaged among all 27 wind scenarios. The results are shown in Table 5.1 and lead to the following conclusions.

a) Even for constant wind $W20_{\text{act}} = 0$, an estimate of $W20_{\text{est}} = 0$ leads to bad results. This can be explained when looking at the computation of the Wind Process noise, for $W20_{\text{est}} = 0$, this results in qwind = 0 , thus not allowing the wind estimate to change at all. So the initial guess will be kept forever? proof
deviation, due to the nonperfect modelling and uncertainties in place. (For perfect modelling this deviation really? or just for constant wind = 0

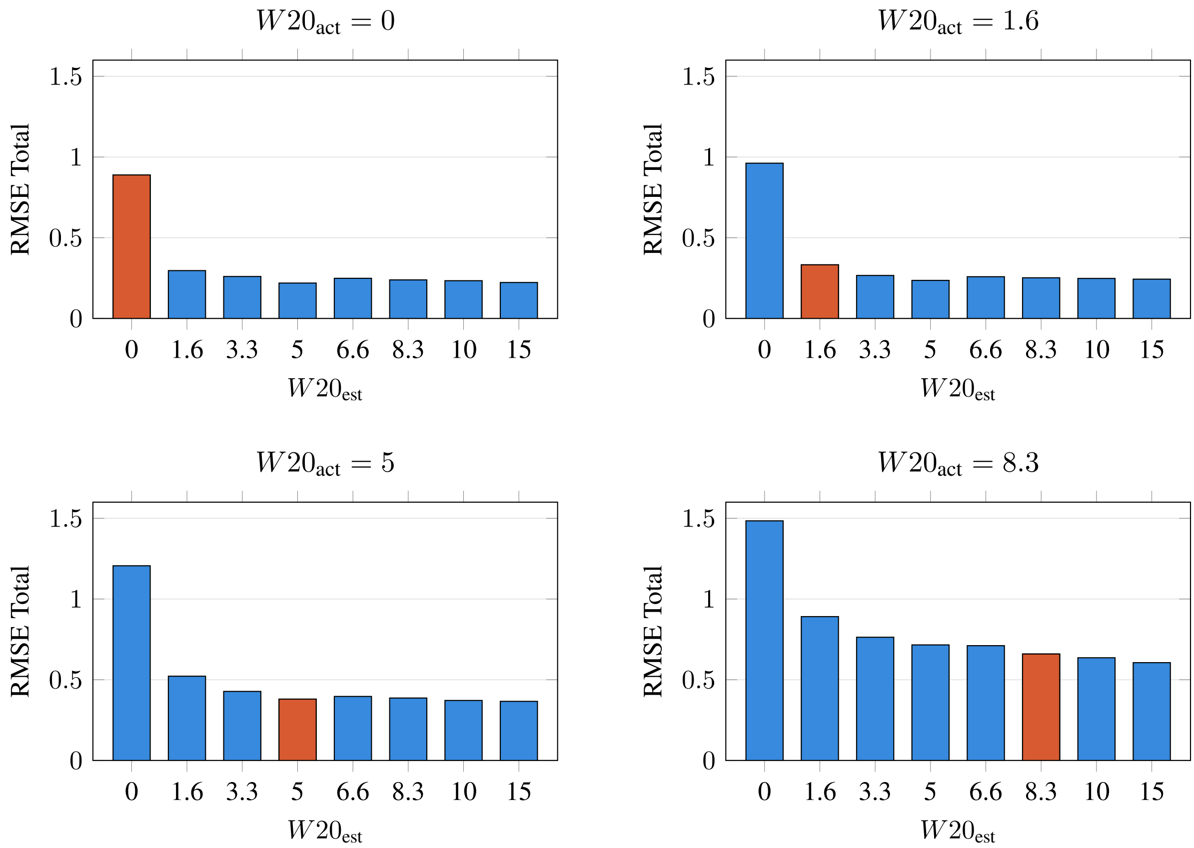

Figure 5.1.: Comparison of Total RMSE Across All Estimated $W20$ Wind Speeds. The orange bar highlights the true configuration scenario where $W20_{\text{est}} = W20_{\text{act}}$.

would be zero) b) matching W20 do not always imply best performance. Reason herefore are errors in modelling and bigger W20 more flexibility to adapt. So why not chosing W20 = 15 for all? Now look at plots showing differences and oscillations.

The results indicate that a despite the real W20 parameter an estimated W20 = 10 yields good results for all real wind conditions created by varying W20. This can be explained as follows: a bigger W20 estimated increases the Process noise, thus giving the wind states more freedom to react to fast changing wind gusts. This comes at a cost which is a more noisy estimation of the wind. W20 estimated = 10 seems to offer a good tradeoff although bigger values may lead to an even lower RMSE. And bigger W20 leads to bigger qwind, this may have bad side-effects.

What happens with the wind estimation if W20est too big? -> if theres deviation in the measurements and the expected behaviour computed by the kalman filter, the error will more likely be attributed to the wind as its more uncertain.

5.1.2. Effect of Airspeed Sensor

The Airship is equipped with an 1-D airspeed sensor. Its interesting to know how the wind estimation changes if no airspeed sensor is in place, to quantify the importance of the AS.

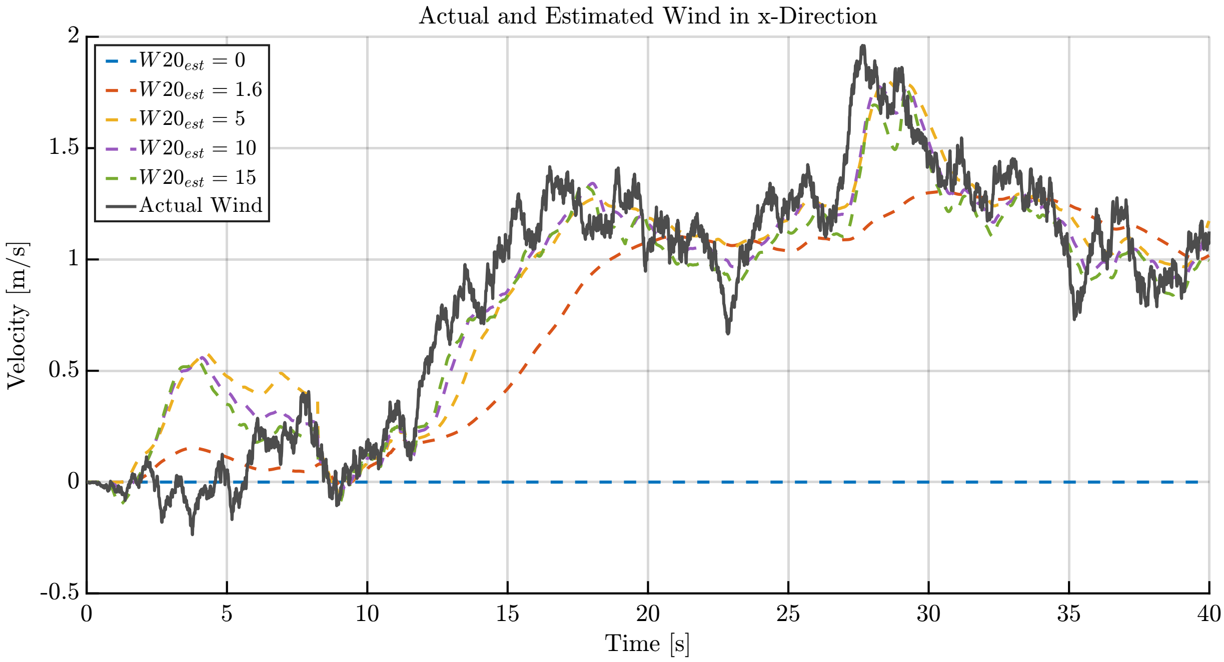

Figure 5.2.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

5.2. Wind compensation

The wind compensation is heavily dependant on the quality of the wind estimation.

Setting:

Results We are looking for one controller which yields the best results for all scenarios. Although we may find better settings for single scenarios we do not want to need to switch.

While theoretically the compensation should allow for a more aggressive PID as the disturbance should compensate the wind disturbance this does not hold up in the results presented. Reason therefore is errors in wind estimation, convergence rates of the KF, maybe even delay of one time step, general uncertainty in the model (uncertain forces etc)

5.2.1. Problems with Observability

Due to the lack of sensory equipment to measure airspeed the wind estimation depends heavily on the effect of the aerodynamic forces on the airship to estimate the wind velocities. Because only the aerodynamic Forces are directly dependent on the airspeed and thus on the wind velocity. The Compensation uses the current wind estimate to compute the estimation and thus requires a good estimation of wind. Exactly there lies the problem, if the compensation fully compensates for the effect of wind on the airship, the wind estimation quality automatically decreases, leading to a poor estimation and so on. To break up this cycle we need to guarantee that the information about the effect of the wind is not fully lost. To achieve this a gain, manipulating the output of the compensation is introduced. This gain in range 0 to 1 determines how aggressively the wind compensation works and balances the wind compensation with the wind estimation.

5.3. Flight Test

6. Conclusion and Outlook

Outlook

The model-based approach proves to be a powerful tool for wind estimation given a good model of the system is available. The Dryden wind model does not only provide a mathematical description for the wind speeds acting at the center of gravity but also yields a description for the effect of the wind on the angular rates, which are caused by differences in wind magnitudes within the prevailing wind field.

Including the angular rates induced by the differences in wind speeds within the wind field can improve the performance as the airship is especially vulnerable to wind fields considering the airship's large side surfaces. Angular rates of the wind are described by transition functions of order 2 (for the u-direction) and order 3 (for v, w direction) resulting in a increase of one order for every direction compared to wind speeds. It remains to investigate whether the gain in performance can justify the increased complexity.

The Wind estimation proved to be particulary reliable. This information can be used for advanced control and trajectory planning algorithms like MPC to increase the performance compared to the current PID setup.

Bibliography

[1] Thor Fossen. *Handbook of Marine Craft Hydrodynamics and Motion Control*. Apr. 2021. ISBN: 978-1119575047. DOI: 10.1002/9781119994138.

[2] Yuwen Li and Meyer Nahon. “Modeling and Simulation of Airship Dynamics”. In: *Journal of Guidance, Control, and Dynamics* 30.6 (2007), pp. 1691–1700. DOI: 10.2514/1.29061. URL: https://doi.org/10.2514/1.29061.

[3] John D. Anderson. *Fundamentals of Aerodynamics*. 6th ed. McGraw-Hill Education, 2017.

[4] J. N. Newman. *Marine Hydrodynamics*. The MIT Press, Aug. 1977. ISBN: 9780262280617. DOI: 10.7551/mitpress/4443.001.0001. URL: https://doi.org/10.7551/mitpress/4443.001.0001.

A. First Appendix

Write your appendix content here.

A.1. Parameters of the airship squid

Table A.1.: General Airship Parameters and Specifications

Parameter	**Symbol**	**Value**	**Unit**
Overall length	L	75	m
Maximum diameter	D	18	m
Envelope volume	V	10 000	m^3
Total mass	m	8 500	kg
Payload mass	m_p	1 500	kg
Helium mass	m_{He}	1 700	kg
Cruise speed	v_c	90	km/h
Maximum speed	$v_{\max}$	120	km/h
Maximum altitude	$h_{\max}$	3 000	m
Endurance	t_e	24	h
Propulsion power	P	2×150	kW
Number of engines	N_e	2	–
Buoyant force	F_b	98 000	N
Drag coefficient	C_D	0.045	–
Reference area	A	255	m^2
Moment of inertia (roll)	I_{xx}	1.2×10^5	kg m^2
Moment of inertia (pitch)	I_{yy}	8.5×10^5	kg m^2
Moment of inertia (yaw)	I_{zz}	8.9×10^5	kg m^2
Center of buoyancy offset	z_B	0.8	m

Table A.2.: Modeling Coefficients and constants

Coefficient	**Symbol**	**Value**	**Unit**
Motor Coefficient	C_{motor}	x	y
alphas	α	x	in N!!!

B. Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.